DSC5001 Project Proposal

Team FraudML · Submitted 5 November 2026

Project title	Can Machine Learning Find More Payment Fraud Under a Fixed Review Budget?
Team name	FraudML

Member	UID	Main responsibility
Wang Yashi	[UID]	System integration and project coordination
Li Yihui	[UID]	Data preparation and exploratory analysis
Wang Shihong	[UID]	Baselines and Logistic Regression
Wu Jiaqi	[UID]	LightGBM and feature comparison
Chen Situ	[UID]	Interpretation and presentation

1. Background and Motivation

Payment reviewers have limited capacity. Payment-system development experience motivates comparing model scores with amount-based prioritization. We measure labeled fraud selected within a fixed review budget.

2. Research Questions

- RQ1 Does model-based ranking find more labeled fraud than amount ranking among 500 selected transactions in a future period?
- RQ2 Do strictly historical card/address statistics improve LightGBM beyond categorical inputs?
- RQ3 How does ranking quality change between validation and test, particularly when extreme inputs produce saturated scores?

3. Data

- Source IEEE-CIS Fraud Detection, provided by Vesta through Kaggle [1], accessed 5 November 2026. TransactionDT provides relative time.
- Unit and size Each row is a transaction. The labeled transaction file has 590,540 rows and 394 columns. The comparison excludes the identity table.
- Response isFraud is a binary historical label.
- Predictors Amount, product, card/address proxies and anonymous attributes. Additional features summarize prior count and mean amount by card1 and addr1.
- Known issues Imbalance, missingness, outliers and unclear upstream definitions. Proxy entities may combine unrelated activity. Anonymous feature availability cannot be audited.

4. Proposed Methods

- Baseline Seeded random ranking and descending amount ranking address RQ1.
- Interpretable model Balanced Logistic Regression addresses RQ1 and RQ3.
- Flexible model LightGBM [2] captures nonlinear relationships. An identical configuration with additional historical statistics addresses RQ2.
- Preprocessing Fit imputation, scaling and encodings on training data. Frequency-encode high-cardinality categories. Exclude transaction ID, outcome and raw split time.

5. Validation and Evaluation

- Split Use chronological 70/15/15 training, validation and test periods, retaining equal timestamps together. Historical features use strictly earlier timestamps and exclude labels.
- Tuning Use fixed estimator settings and seed 42, with configuration comparison on validation. Holdout comparisons are complete. Further changes use validation, and test diagnostics are identified as post-hoc.
- Metrics Report average precision [3], Precision@K, Recall@K and labeled fraud found at K=500, with K=100 as sensitivity analysis. K applies to the period rather than each day.
- Interpretation and robustness Compare period outcomes, errors, standardized-feature extremes and LR probability versus decision-score rankings. Single-seed results have no confidence intervals and support baseline-level conclusions.

6. Expected Contribution

We evaluate investigation yield at fixed capacity and the incremental value of historical features. A prototype links evaluated Top-K selection to an auditable queue, including high-score cases. Scores are uncalibrated ranking signals. Dataset anonymity, one holdout and simulated review operations limit conclusions about deployed banking performance.

7. Timeline

Final submission is due 29 November 2026 at 11:59 PM. Internal completion is 25 November.

Week	Dates	Milestone
1	5-8 Nov	Audit data, preserve existing results and finalize proposal
2	9-15 Nov	Verify baseline implementation, features and capacity-based review
3	16-22 Nov	Complete interpretation and robustness analysis, freeze report
4	23-29 Nov	Complete by 25 Nov, prepare at most 10 slides, 8-minute recording and code appendix, submit by 29 Nov

References

[1] Kaggle. IEEE-CIS Fraud Detection. 2019. https://www.kaggle.com/competitions/ieee-fraud-detection. Accessed 5 November 2026.

[2] Ke, G., et al. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. NeurIPS, 2017.

[3] Saito, T., and Rehmsmeier, M. The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. PLOS ONE, 10(3), e0118432, 2015.